

820-0229 – Install instructions for PN 205-0121-01 Pump (GE PN 5156321)

PPE Required: Safety glasses, Safety shoes, Gloves

Tools Required:

- Strap wrench (supplied with FRU)
- Hoist Service Kit 5196226
- Non-Magnetic Tool Kit 5113258
- Bucket
- DVMR Coolant 5174313-4 (1 kit = 4 US Gal)

Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Disable MagMon from sending alarms during procedure.
3. Lockout/Tagout system in accordance with GE procedures.
4. Drain the respective pump by:
 - a. Closing the gate valve beneath the reservoir,
 - b. Direct the hose at the bottom of the pump into bucket.
 - c. Open the drain valve and remove the bolt on the face of the pump 180° from the drain.

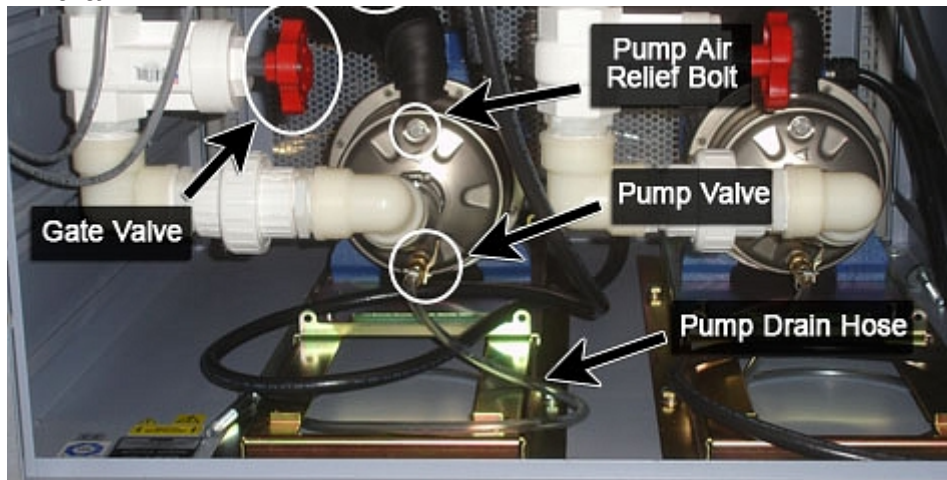


Photo 1 Component Locations for Draining

5. Install Hoist Assembly (reference GE service procedure) and Chain lift assembly included in Pump FRU kit.
6. Open Front cover of Power box inside HEC.
7. Locate Variable Frequency Drive (VFD) for pump in question..
8. Remove pump supply wiring from VFD (record wire locations for re-installation).
9. Remove power cord from bottom of Electrical Box.
10. Remove discharge hose from top of pump using Slip-joint pliers on SAE flare fitting.

11. Break suction hose from tank at union (see Photo #2) using strap wrench



Photo 2 - Breaking the suction hose connection to the tank.

12. Loosen retaining screws for pump tray (2 places – Refer to photo)

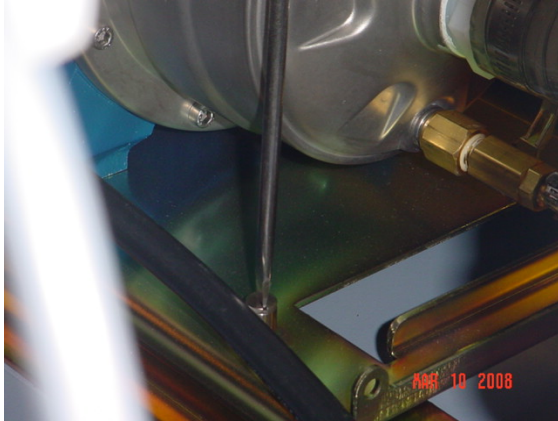


Photo 3 Loosening the pump tray retain screws.

13. Grab tray by built in handle and pull forward until it stops. Lift up over lip of cabinet to facilitate lifting with hoist.
14. Assemble Hoist Kit to HEC.
15. Attach Chain lift assembly to pump lifting brackets on 4 corners of pump plate. Ensure that both chains going to back brackets are mated to the same link. Ensure the front bracket chains are likewise mated. This will provide a balanced lift.
16. Attach Chain lift assembly to Hoist assembly.



Photo 4 Hoist Assembly Hook and Chain detail.

17. Place Hoist Assembly handle in “Up” position.



Photo 5 Hoist Handle in “UP” position.

18. Ratchet Hoist Assembly handle to begin lifting the pump.

19. When pump is high enough to clear cabinet, carefully swing Chain lift assembly toward the packing crate.

20. When pump is in proper position, place Hoist Assembly ratchet in “Down” position.

21. Lower pump until it is settled in crate.

End of Procedure

Installation Procedure

1. Machine should be in a shutdown condition as required by above procedure.
2. Position Pump crate as close to machine as possible to facilitate lifting of the pump.
3. Attach Hoist Assembly chain hook to the pump through 4 eyelets located on the pump tray.

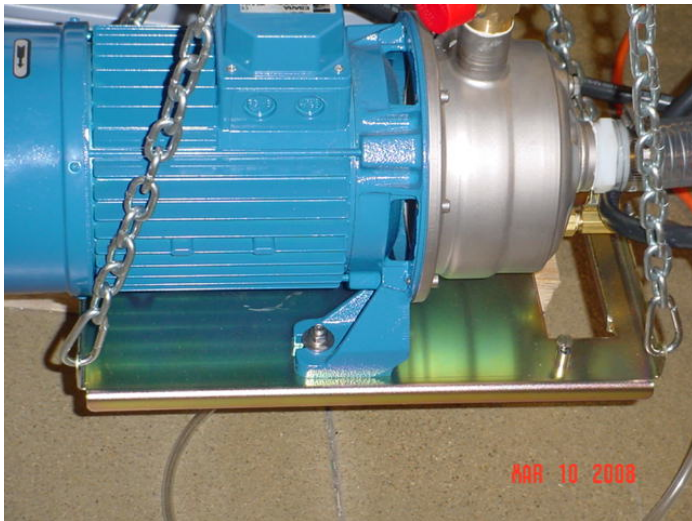


Photo 6 Detail of lifting the pump assembly

4. Place Hoist Assembly handle in the "UP" position.
5. Take slack out of chain length by ratcheting the handle.
6. While guiding the tightened chain with your free hand, slowly lift the pump until it clears the lip of the machine.
7. Slide the pump into the machine along the Hoist Assembly track.

NOTE: The pump will be unable to go the complete distance due to the angle with the machine.

8. Lower the pump until the pump front end rests on the lip of the machine.
9. Remove the Hoist Assembly chain hooks from the pump.
10. Move the pump onto the pump positioning tracks.
11. Slide the pump back into the machine and align the two thumbscrews that secure the pump.
12. Remove the Hoist Assembly from the machine.
13. Reattach discharge hose at top of pump.

NOTE: Depending on the revision of the HEC, it is possible that there is a mismatch between pump suction hoses. The molded blue hose should be used whenever possible.

To install the molded hose:

1. Remove existing pump suction hose assembly from HEC (leave gate valve in place: ensure it does not rotate when removing fittings with slip-joint pliers)



2. Wrap threads of supplied plastic barbed fittings with the supplied PTFE tape. Wrap entire area of threads with two full wraps (in a clockwise direction when looking at the threaded end).
3. Thread new hose barb fitting into bottom of gate valve to make a leak-tight connection.



4. Place the supplied Breeze-brand hose clamps over pump suction hose and install over barb on front of the pump and beneath the gate valve (hose is orientation specific and can only be installed one way). Refer to photo below



5. Tighten Breeze hose clamp until the stacked washers flatten.

14. Fill the system in accordance with Lytron manual #820-0190. Ensure gate valve beneath tank is fully open.
15. Route the power cable for the pump through the machine to the Electrical Box.
16. Wire the pump to the appropriate Variable Frequency Drive (VFD). Wires are phase sensitive and need to be installed in correct order (see labels on wires)
17. Clear the Lockout/Tagout in accordance with GE procedures.
18. Re-start HEC and turn pump ON. Add new DVMR coolant as necessary.
19. Check for any leakage. Repair if necessary.

End of Procedure